AOOP ASSIGNMENT WEEK-1

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Sec-4

1. Create a class Student with student ID, name and age. Create three student objects and display their details.

```
class Student {
   int id;
   String name;
   int age;

   void display() {
       System.out.println("Student ID: " + id);
       System.out.println("Name: " + name);
       System.out.println("Age: " + age);
   }

   public static void main(String[] args) {

       Student s1 = new Student();
       s1.id = 101;
       s1.name = "Riya";
       s1.age = 18;

       Student s2 = new Student();
       s2.id = 102;
       s2.name = "Priya";
       s2.age = 19;

       Student s3 = new Student();
       s3.id = 103;
       s3.name = "Anu";
       s3.age = 20;

       s1.display();
       s2.display();
       s3.display();
   }
}
```

Output:

```
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % javac Student.java
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % java Student
Student ID: 101
Name: Riya
Age: 18
Student ID: 102
Name: Priya
Age: 19
Student ID: 103
Name: Anu
Age: 20
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 %
```

2. Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

```
class Employee {
   int id;
   String name;
   double salary;

   Employee(int i, String n, double s) {
       id = i;

       name = n;
       salary = s;
   }

   void display() {
       System.out.println("Employee ID: " + id);
       System.out.println("Name: " + name);
       System.out.println("Salary: " + salary);
   }

   public static void main(String[] args) {

       Employee e1 = new Employee(101, "Riya", 30000);
```

```
        e1.display();
    }
}
```

Output:

```
  Age: 20
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % javac Employee.java
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % java Employee
  Employee ID: 101
  Name: Riya
  Salary: 30000.0
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 %
```

3. Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.

```
class BankAccount {
    int accountNumber;
    double balance;

    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
```

```
        BankAccount b1 = new BankAccount();

        b1.accountNumber = 12345;
        b1.balance = 5000;

        b1.deposit(2000);
        b1.withdraw(1000);

        b1.display();
    }
}
```

output:

```
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % javac BankAccount.java
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % java BankAccount
Deposited: 2000.0
Withdrawn: 1000.0
Account Number: 12345
Balance: 6000.0
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 %
```

4. Create a Rectangle class with length and width. Calculate and display its area and perimeter.

```
class Rectangle {
    int length;
    int width;

    void calculate() {
        int area = length * width;
        int perimeter = 2 * (length + width);

        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {

        Rectangle r1 = new Rectangle();

        r1.length = 10;
```

```
        r1.width = 5;

        r1.calculate();
    }
}
```

output:

```
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % javac Rectangle.java
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % java Rectangle
Area: 50
Perimeter: 30
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 %
```

5. Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

```
class Book {
    int isbn;
    String title;
    double price;

    Book(int i, String t, double p) {
        isbn = i;
        title = t;
        price = p;
    }

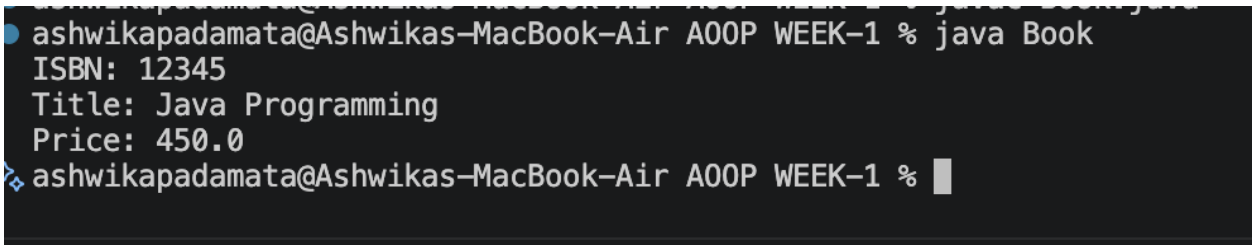
    void display() {
        System.out.println("ISBN: " + isbn);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {

        Book b1 = new Book(12345, "Java Programming", 450);

        b1.display();
    }
}
```

output


ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 % java Book
ISBN: 12345
Title: Java Programming
Price: 450.0
ashwikapadamata@Ashwikas-MacBook-Air AOOP WEEK-1 %

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